

Payments Architecture — Go / No-Go

Internal reference document — CEO walkthrough, May 2026

01 · THE PROBLEM

Every payment Truemeds processes runs through one gateway.

Cashfree handles 100% of our transactions. If it's slow, our checkout is slow. If it's down, revenue stops — no fallback, no reroute, no negotiating leverage.

GMV	Transactions / month	Payment SR · Apr 2026	Avg order value
₹35.8 Cr	2.81L	83.26%	₹1,274
April 2026		83.05% in March	

Single point of failure

Any Cashfree outage — partial or full — hits revenue directly with zero fallback. This is the architecture problem we're solving.

Why the answer is multi-PG, not a PG swap

Reason	What it means
Zero-downtime fallback	If any one PG is slow or down, traffic reroutes to another instantly. Revenue doesn't stop.
Negotiation leverage	PGs compete on rates when they know you can switch. April data proves this — PayU is ₹1.07L/month cheaper than Cashfree on the same transactions.
SR competition	PGs optimise their Success Rate (SR — the share of payment attempts that complete) harder when routing isn't guaranteed. Today Cashfree has no reason to.

Routing control

Route each payment instrument to the PG with the best SR for it that month. Right now every transaction goes to Cashfree — no choice.

Four options. Two made it through.

Two paths were ruled out before any vendor conversation.

Option	Status	Full reasoning
Stay on a Single PG	✗ Eliminated	You still have one gateway. Any outage = full stop, no fallback. If that PG is Razorpay — the only other tier-1 option — the picture gets worse: most expensive PG in our comparison at ₹34.71L/month, and their SR optimisation feature (Edge Routing) doesn't work on Truemed's payment link flow. You'd be paying the premium without the feature it's sold on. Same architecture problem, different vendor name, higher cost.
Build In-House	✗ Eliminated	18–24 months, 8–10 engineers, uncertain outcome. Scope: Payment Card Industry Data Security Standard (PCI-DSS) compliance, individual bank integrations for net banking, tokenisation, reconciliation engine, retry logic, fraud detection — all owned and maintained by us permanently. Even if built, we'd need PG relationships underneath — effectively becoming Juspay with none of Juspay's advantage: 300M+ daily transactions, pre-emptive bank rail failure detection, and years of routing intelligence we'd start from zero on.
Razorpay Optimizer	→ Evaluated	PG-owned orchestration with multi-PG support. Evaluated solely because Razorpay won't join Juspay.
Juspay + Multi-PG	→ Evaluated	Neutral orchestration above multiple PGs. Architecture-first choice.

Two very different architectures.

Path A — Razorpay Optimizer

Multi-PG routing and retry — with Razorpay's interests baked into the logic.

Why it's on the table

Razorpay doesn't support Juspay. Confirmed by both vendors. That's the only reason this path is being evaluated.

Industry context — the Juspay scare

In 2024, PGs paused Juspay integrations — Juspay makes PG-switching easy, driving price competition against them. By mid-2025, Cashfree, PayU, PhonePe, and EaseBuzz had all returned. Razorpay is the only holdout. This is a commercial call, not a technical one — Razorpay is protecting its pricing power. The fact that every other PG returned is evidence that Juspay works as intended.

What Razorpay Optimizer offers

Routing rules, retries, and analytics across Razorpay and select PGs. Routing priority favours Razorpay's own rails — Razorpay retries its own instruments first before switching to another PG.

Why Edge Routing doesn't work for Truemedics

Problem	Explanation
1 — Our flow vs. what Edge Routing needs	Truemedics generates payments as follows: our backend creates a payment link → link sent to customer → customer taps it and it opens in their browser or app screen. Edge Routing is designed to work when Razorpay's SDK is embedded directly inside your app — it reads device signals, session data, and network conditions in real time. When a customer opens a payment link in a browser or WebView, that SDK context doesn't exist. Edge Routing never activates.
2 — Razorpay down = links stop generating entirely	Before Truemedics can create a payment link, our backend calls Razorpay's servers to get a session token. That's step one. If Razorpay's servers are slow or down, that call fails. No token = no link = customer never sees a payment page. The failure happens before Edge Routing — or any Razorpay feature — can do anything.
3 — Retry still means Razorpay only	If a payment fails, Optimizer retries — but only within Razorpay's own instruments: different card network, different bank, different UPI handle. All still Razorpay. There is no "try PayU instead." If Razorpay has an issue with a specific bank, every retry hits the same wall.

Cost

Already the most expensive PG in the comparison at ₹34.71L/month on April's transaction mix. Optimizer adds further cost on top (fee not publicly quoted).

Path B — Juspay + Multi-PG

Neutral layer above multiple PGs — Truemed's controls the routing, not the PG.

Where Juspay operates across the payment funnel

Funnel stage	What Juspay does
Checkout UI	HyperCheckout — pre-filled saved cards, UPI intents surfaced upfront before the customer selects manually.
Saved cards	Juspay Safe — cross-PG tokenisation (Payment Card Industry Data Security Standard / PCI-DSS compliant). A card saved while routing through PayU works when routing through Cashfree. No instrument dropout on PG switch.
Amount hold	UPI Block / Dynamic Payable — holds the payment authorisation for the correct final order amount. Truemed's orders change post-placement (substitutions, doctor or health advisor interventions). Juspay handles this natively, PG-agnostic. PayU has also confirmed this feature.
Routing	Cost-first, SR-first, or a weighted combination — configurable at instrument and issuer level. Example: route HDFC credit cards via whichever PG has the best HDFC SR that month.
Retry	Many Truemed's customers are from smaller cities — patchy internet and bank transaction blocks mid-payment are common. Juspay's retry engine recovers those transactions automatically by rerouting through a different instrument or gateway instead of showing a failure screen. Powered by pre-emptive bank rail failure detection from 300M+ daily transactions across the Juspay network.
Offer engine	Centralised coupon layer above all PGs. One coupon configured once works regardless of which PG processes the transaction. Without this, every PG integration needs its own offer logic built and maintained separately — engineering effort that compounds with each new PG added.
Post-payment	Unified webhook normalisation + reconciliation across all PGs. One integration point for finance and ops regardless of which PG settled the transaction.

One limitation

Fresh integration required from scratch — same as Razorpay. Razorpay Optimizer requires more: PG integration plus additional frontend SDK changes for Edge Routing to work at all. Juspay's HyperCheckout is the integration, not an add-on.

PGs confirmed on Juspay platform: Cashfree, PayU, EaseBuzz, PhonePe — all active.

What we spend today. What a move costs.

April 2026 — Payment Mix by GMV

₹35.84 Cr across 2.81L transactions

Payment mode	% of GMV	GMV (approx.)
UPI (all variants)	71.3%	₹25.55 Cr
of which: Pure UPI (0% MDR)	60.4%	₹21.66 Cr
Credit Card (incl. UPI CC)	23.6%	₹8.45 Cr
Wallets & Others	2.6%	₹0.93 Cr
Debit Card	1.9%	₹0.68 Cr
Net Banking	0.6%	₹0.21 Cr

PG Cost Comparison — Same April Transaction Mix

Each PG's rate card applied to Truemed's actual April data.

PG	Monthly cost	vs Cashfree	Note
PayU	₹24.31L	−₹1.07L	Cheapest option
EaseBuzz	₹24.80L	−₹0.58L	
Cashfree (actual)	₹25.38L	Current	Today's baseline
Razorpay	₹34.71L	+₹9.33L	Most expensive

UPI is 60.4% of GMV (₹21.66 Cr/month) — Razorpay is the only PG that charges for it

Razorpay offers a 0.20% UPI tech fee as an optional SR upsell. Every other PG charges 0% on UPI. If taken: ₹5.11L/month on transactions that cost nothing elsewhere. What's included in that fee: • SR flag enablement — activates Razorpay's UPI SR optimisation layer • SR-sensitive user base targeting — routes higher-risk UPI transactions through better-performing paths • Dedicated analytics POC — a Razorpay resource who monitors SR at payment mode and acquirer level Their claim: UPI SR (Success Rate) lifts from ~94-95% to ~97% with this enabled. Honest view: The claimed improvement is at the UPI gateway level only — it measures uptime and routing performance within the payment step. It does not address the overall payment funnel where customer intent, network conditions, and external factors determine whether a transaction completes. These are different problems.

Juspay — What It Adds

+₹4.85L / month over Cashfree today

PayU (cheapest PG) + Juspay = ₹30.23L/month vs ₹25.38L on Cashfree today.

Break-even at modest SR improvement — see calculator below.

Item	Value
Juspay pricing (confirmed quote)	14 bps on GMV
April GMV	₹35.84 Cr
Juspay monthly cost (incl. 18% GST)	₹5.92L
Best-priced PG today (PayU) + Juspay	₹30.23L
vs Cashfree today	+₹4.85L / month incremental

SR Break-even Calculator — Contribution Margin: 30%

Minimum SR lift to cover Juspay cost in the same month: **0.46%** Juspay costs ₹5.92L/month. At 30% contribution margin, the table below shows what each SR improvement puts back.

SR Lift	Additional GMV / month	Contribution recovered (30%)	vs Juspay cost (₹5.92L)
0.5%	₹21.52L	₹6.46L	✓ Covers Juspay cost
1.0%	₹43.04L	₹12.91L	✓ Covers Juspay cost
1.5%	₹64.57L	₹19.37L	✓ Covers Juspay cost
2.0%	₹86.09L	₹25.83L	✓ Covers Juspay cost
2.5%	₹1.08 Cr	₹32.28L	✓ Covers Juspay cost
3.0%	₹1.29 Cr	₹38.74L	✓ Covers Juspay cost

Juspay + Multi-PG.

	Razorpay Optimizer	Juspay + Multi-PG
Orchestration model	PG-owned — Razorpay's rails get priority. Routing logic controlled by Razorpay.	Neutral — Truemeds sets the routing rules. No PG has preference.
Why this matters	Razorpay has a financial stake in which PG wins. Their routing optimises for their rails, not necessarily Truemeds' best outcome.	Juspay earns the same regardless of which PG processes. True neutrality.
PG coverage	Razorpay + select PGs (e.g., PayU)	Cashfree, PayU, EaseBuzz, PhonePe
Funnel coverage	Routing + retry confirmed. Full checkout coverage — not verified.	Checkout → reconciliation (full funnel confirmed)
Monthly cost	₹34.71L base + Optimizer fee (unquoted)	₹30.23L (PayU + Juspay)
Integration effort	Fresh build + frontend SDK for Edge Routing (more work)	Fresh build — HyperCheckout is the integration
Offer engine	Not confirmed	Centralised across all PGs — out of the box
Dynamic Payable / UPI Block	Not confirmed	Confirmed, PG-agnostic
Why evaluated	Razorpay won't join Juspay	Architecture-first choice

Recommended: Juspay + Multi-PG

Removes single-PG dependency. Covers the full payment funnel — not just routing. The ₹4.85L/month incremental over Cashfree today is recoverable at modest SR improvement.

Razorpay Optimizer is evaluated for one reason: Razorpay's commercial decision to stay off Juspay. If that reverses, they join Juspay's network — and the Optimizer question becomes irrelevant.

Integration reality

Both paths require a fresh integration from scratch — Truemeds is not on Razorpay today, no head start either way. Razorpay Optimizer requires more: PG integration plus additional frontend SDK changes for Edge Routing to work at all. Juspay's HyperCheckout is the integration — not an add-on to one.

Three phases. 12 weeks to full go-live.

The ask

Approve the Juspay + Multi-PG path and unblock the Payments POD to begin vendor onboarding and integration planning.

What we're actually building

Three components, in sequence.

Component	What it is	Who
Juspay integration	HyperCheckout + HyperRouter — checkout UI, routing engine, retry logic	Payments POD + Engineering
PG onboarding	PayU as primary (cheapest in April). Cashfree retained as live fallback. Third PG (EaseBuzz or PhonePe) added in Phase 3 once routing is stable.	Payments POD + Finance
Reconciliation layer	Unified webhook normalisation across PGs, finance ops alignment	Engineering + Finance

Phased rollout

2 engineers as primary workload. Third engineer available as buffer. Both paths require fresh integration — this is not a Razorpay advantage.

Phase 1 — Integration & Shadow Mode

Weeks 1–8

Juspay integrated. PayU onboarded. Routing rules configured. Running in shadow alongside live Cashfree — no real traffic moved yet. Validate SR data and routing logic before any cutover. 8 weeks (not 6) to account for Truemed's payment link architecture complexity.

Phase 2 — Controlled Rollout

Weeks 9–12

10% → 30% → 100% traffic migration. Cashfree stays live as hard fallback throughout. Watch SR delta, reconciliation accuracy, and failure patterns at each step.

Phase 3 — Full Go-Live + Optimisation

Week 12+

100% traffic on Juspay + multi-PG. Instrument-level routing tuning begins. Add third PG (EaseBuzz or PhonePe) once routing is stable and SR data justifies it.

What success looks like at 90 days

- ✓ Juspay live on 100% of transactions
- ✓ SR delta measured and tracked against pre-migration baseline — break-even math from Section 4 validated in production
- ✓ Reconciliation running cleanly across both PGs
- ✓ No single-PG dependency — Cashfree as fallback, not primary

Two risks to track

SR lift below model: If improvement is under 0.5% after 60 days on full traffic, revisit the cost case. The break-even calculator in Section 4 shows the floor. Timeline slip: Phase 1 is buffered at 8 weeks for Truemed's payment link complexity. If it runs longer, Cashfree fallback means zero revenue risk while we wait.